

FUNCTIONAL REQUIREMENTS DOCUMENT

DairyPro 360 — All-in-One Dairy Management System

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1. Purpose & Scope

This Functional Requirements Document (FRD) translates the validated business requirements (BRD-DMS-001 v1.1) into detailed, testable functional specifications. Each requirement is expressed as a user story with acceptance criteria, input/output definitions, and business rules. This document is the primary input for the development team and quality analyst.

2. Module: Authentication & User Management

2.1 User Stories

Story ID	As a...	I want to...	So that...
US-AU-01	System Admin	Create user accounts with roles	New staff can access the system
US-AU-02	All Users	Log in with email + password	I can access my role-specific features
US-AU-03	System Admin	Enable 2FA for admin accounts	Admin accounts are protected
US-AU-04	System Admin	Deactivate a user account	Departed staff lose access immediately
US-AU-05	All Users	Reset my password via email OTP	I can regain access if locked out
US-AU-06	System Admin	View audit log of all user actions	I can investigate any suspicious activity

2.2 Acceptance Criteria — US-AU-02

- Given a registered user, when they enter a valid email and password, then the system returns a JWT token and redirects to the role-specific dashboard within 2 seconds.
- Given an incorrect password, when the user tries to log in, then the system shows 'Invalid credentials' and does not expose whether the email or password is wrong.
- Given 5 consecutive failed login attempts, when the 6th attempt is made, then the account is locked for 15 minutes and an email alert is sent to the admin.
- Given a valid JWT token, when it expires after 8 hours, then the user is redirected to the login page.

2.3 RBAC Matrix

Module	Admin	Farm Mgr	Collection Agent	Finance	HR	Driver	QA
Farm & Cattle	R/W/D	R/W	R	R	-	-	R
Milk Collection	R/W/D	R/W	R/W	R	-	-	R
Processing	R/W/D	R/W	R	R	-	-	R/W
Inventory	R/W/D	R/W	R	R/W	-	-	R
Distribution	R/W/D	R	-	R	-	R/W	R
Billing/Finance	R/W/D	-	-	R/W	-	-	R
HR/Payroll	R/W/D	-	-	R	R/W	-	-
Reports	R/W/D	R	R	R	R	-	R
User Management	R/W/D	-	-	-	-	-	-

R=Read, W=Write, D=Delete, -=No Access

3. Module: Milk Collection & Quality Testing

3.1 Data Flow

Collection Agent arrives at farm → Weighs milk using connected scale → Enters or imports fat%, SNF%, CLR, temperature → System validates quality thresholds → If passed: creates collection record and calculates payment → If failed: flags as 'Rejected', prevents entry to processing, and sends SMS alert → End of day: system generates collection summary.

3.2 Functional Requirements

FR-ID	Requirement	Input	Output	Business Rule
FR-MC-01	Record collection per farm	Farm ID, shift, quantity (L)	Collection record (fat%, SNF%, CLR, Temp)	Simple Batch ID. System auto-timestamps.
FR-MC-02	Quality threshold validation	Fat%, SNF%, temperature	Pass/Reject flag	Fat < 3.0%: Reject. SNF < 8.0%: Reject. Temp > 10°C: Reject.
FR-MC-03	Payment calculation	Quantity, fat%, base rate	Payment amount (INR)	Rate = Base INR 27/L + INR 0.5 per 0.1% fat above 3.0%
FR-MC-04	Daily collection summary	Date, farmer list	PDF/Excel report	Auto-generated at 23:59 daily
FR-MC-05	CSV import from analyser	CSV file from milk analyser	Bulk collection records	System validates each row; rejects invalid rows with error
FR-MC-06	Offline entry mode	Collection data (offline)	Sync queue	Data stored locally; synced when internet restored

3.3 User Story — Payment Calculation

US-MC-03: As a Collection Agent, I want the system to automatically calculate the payment for each farmer based on quantity, fat%, and the rate slab, so that I do not need to do manual calculations.

Acceptance Criteria:

- Given farmer delivers 20L at fat% 4.5: Payment = 20 × (27 + (0.5 × 5)) = 20 × 29.50 = INR 590.
- Given farmer delivers milk with fat% 3.5: Payment = 20 × (27 + (0.5 × -5)) = 20 × 24.50 = INR 490.
- Given fat% < 3.0: Collection is rejected; payment = INR 0; rejection SMS sent to farmer.
- Payment record is linked to the collection batch ID and visible to Finance role.

4. Module: Farm & Cattle Management

4.1 Entity: Cattle

Field	Type	Validation	Description
cattle_id	VARCHAR(20)	Unique, Required	System-generated or scanned tag ID
name	VARCHAR(100)	Optional	Name or label for the animal
breed	ENUM	Required	Holstein, Jersey, Sahiwal, Gir, HF, Other
date_of_birth	DATE	Required, Past date	For age calculation
gender	ENUM	Required	Male / Female
health_status	ENUM	Required	Healthy / Sick / Quarantined / Sold / Deceased
last_vaccination	DATE	Optional	Date of last vaccination
daily_yield_L	DECIMAL(5,2)	Optional, >= 0	Average daily milk yield in litres
notes	TEXT	Optional	Vet notes, special conditions

4.2 Functional Requirements

- FR-FM-01: System shall support full CRUD operations on cattle records with audit trail.
- FR-FM-02: System shall send email/SMS alert when a cattle's health_status changes to 'Sick' or 'Quarantined'.
- FR-FM-03: System shall auto-calculate age from date_of_birth and show in cattle profile.
- FR-FM-04: System shall track daily yield per cattle per session and show 30-day yield trend chart.
- FR-FM-05: System shall schedule vaccination reminders 7 days before due date.
- FR-FM-06: System shall allow bulk cattle import via CSV template.

5. Module: Processing & Production

5.1 Production Batch Workflow

Raw milk from inventory → Supervisor creates production batch → Selects product type (pasteurised milk / curd / ghee / butter / paneer / cheese) → Enters input quantity → System deducts from raw milk inventory → Production proceeds → Supervisor enters output quantity and by-product quantities → System calculates yield efficiency → Batch is closed and finished goods are added to inventory with batch number and expiry date.

5.2 Functional Requirements

FR-ID	Requirement	Business Rule
FR-PP-01	Create production batch with input/output logging	Batch ID auto-generated. Input deducted from raw milk stock.
FR-PP-02	Support multiple product types per batch	One batch can produce pasteurised milk AND curd from same input.
FR-PP-03	Calculate yield efficiency	$\text{Efficiency} = (\text{Total output kg} / \text{Total input L}) \times 100$
FR-PP-04	Track equipment used per batch	Equipment log records machine ID, start time, end time.
FR-PP-05	FIFO enforcement for raw material	Oldest collection batch used first. System enforces; cannot override without admin approval.
FR-PP-06	Alert on yield efficiency < 80%	Email alert to Processing Supervisor and Production Manager.

6. Module: Distribution & Delivery

6.1 Functional Requirements

FR-ID	Requirement	Details
FR-DD-01	Create and assign delivery routes	Route has: vehicle, driver, list of customers, order of stops
FR-DD-02	Generate daily delivery schedule	Auto-generated at 6:00 AM based on pending orders
FR-DD-03	Track delivery status	Status: Scheduled > Dispatched > In-Transit > Delivered / Failed
FR-DD-04	Record proof of delivery	OTP sent to customer; agent enters OTP to confirm delivery
FR-DD-05	Handle returns and rejections	Rejection reason codes: damaged, short qty, wrong product, refused
FR-DD-06	Log trip fuel and cost	Driver enters fuel filled (L) and cost; system calculates cost/km

7. Module: Sales, Billing & Finance

7.1 Invoice Generation Rules

- Invoice auto-generated on order dispatch.
- Invoice includes: customer GSTIN, seller GSTIN, HSN codes (0402 for dairy), CGST/SGST/IGST as applicable.
- Invoice numbering format: INV-YYYY-MM-NNNN (e.g., INV-2025-06-0001).
- Credit note generated on return acceptance.
- Overdue invoices (> 30 days) trigger automated email reminder.
- Payment receipt generated on payment collection and linked to invoice.

7.2 Farmer Payment Rules

- Farmer payment calculated from collection records for the billing period (1st–15th, 16th–31st).
- Deductions applicable: advance recovered, milk rejection penalties.
- Payment advice sent to farmer via SMS with breakdown.
- Payment record linked to collection batch IDs for full traceability.

8. Module: HR & Payroll

8.1 Payroll Calculation Logic

Monthly payroll is calculated on the last working day. The calculation flow is: Gross Salary = Basic + HRA + Special Allowance. Deductions = PF (12% of Basic) + ESI (if gross < INR 21,000: 0.75% employee share) + TDS (as per tax slab) + Advance Recovery. Net Pay = Gross - Deductions. Payslip generated as PDF and emailed to employee.

8.2 Functional Requirements

- FR-HR-01: System shall maintain employee master: name, designation, department, joining date, salary structure, bank account.
- FR-HR-02: System shall support attendance marking via OTP (for field staff) and web login (for office staff).
- FR-HR-03: System shall auto-calculate monthly payroll with PF, ESI, TDS deductions.
- FR-HR-04: System shall allow HR Manager to apply manual adjustments before final payroll run.
- FR-HR-05: System shall generate payslip PDF per employee and send via email.
- FR-HR-06: System shall track leave balance (CL, PL, SL) and reduce on approved leave.

9. Module: Reports & Analytics

Report Name	Audience	Frequency	Key Metrics
Executive KPI Dashboard	Dairy Owner	Real-time	Total milk collected, Revenue, Active cattle, Deliveries
Daily Collection Report	Farm Mgr, Collection Agent	Daily	Farmer-wise volume, fat%, payment due
Production Efficiency Report	Processing Supervisor	Daily	Batch yield %, downtime, product output
Inventory Status Report	Warehouse Manager	Daily	Stock levels, expiry alerts, reorder status
Delivery Performance Report	Distribution Manager	Daily	On-time %, returns %, fuel cost/trip
Revenue & P&L Summary	Finance Officer	Monthly	Revenue by product, COGS, gross margin
Farmer Contribution Report	Finance, BA	Monthly	Farmer-wise volume, quality, payment history
Cattle Performance Report	Farm Manager	Monthly	Per-cattle yield trend, health events
Demand Forecast	Owner, PM	Monthly	3-month moving average sales forecast by SKU

10. Module: Inventory Management

10.1 Functional Requirements

- FR-IN-01: System shall track raw material stock (raw milk, packaging, additives) by warehouse/cold storage.
- FR-IN-02: System shall track finished goods by SKU, batch number, production date, and expiry date.
- FR-IN-03: Stock is automatically deducted on production start and delivery dispatch.
- FR-IN-04: Reorder alert triggered when stock falls below minimum reorder level (configurable per item).
- FR-IN-05: System shall support multi-location inventory with transfer records between locations.
- FR-IN-06: Stock valuation report uses weighted average cost method.

11. API Contract Summary

The backend will expose a REST API. Key endpoint patterns:

Endpoint	Method	Description
/api/auth/login	POST	Authenticate user, return JWT
/api/cattle	GET/POST	List cattle / Create new cattle record
/api/cattle/{id}	GET/PUT/DELETE	Get, update, or delete a cattle record
/api/collection	GET/POST	List collections / Record new collection
/api/collection/import	POST	Bulk import from CSV
/api/production/batch	GET/POST	List batches / Create production batch
/api/inventory/{type}	GET	Get current stock levels (raw/finished)
/api/delivery/schedule	GET/POST	Get today's schedule / Create delivery

/api/invoices	GET/POST	List invoices / Generate invoice
/api/payroll/run	POST	Trigger monthly payroll calculation
/api/reports/{type}	GET	Generate and return report data